STATUS · 2 NOVEMBER

Where the project stands

W1	1 · Adaptive lane following	DRIVEN 60 S
W1	2 · Obstacle detection & stop	8 STOPS LIVE
W1	2 · Obstacle bypass maneuver	MOVES TESTED
W3	6 · State machine	6 STATES RUN
W2	3, 4, 5 · Signs, intersection, parking	BLOCKED ON TRACK SETUP
W4	7 · Distance-based cruise control	BLOCKED ON 2ND ROBOT
W5	8 · Final run	OPEN

Week 1 complete. Green: driven on the track. Amber: code exists, no complete run. Grey: blocked on something physical.

What we did since the last demonstration

26 OCT · FIRST AVOID RUN

Failed

Premature lane re-engagement resulting in orthogonal track departure.

26 OCT · FIX

Camera ends the pass

The pass now ends when the cup mask stops seeing the cup. Recovery needs two tape lines.

29 OCT · SECOND RUN

Three passes, wrong angle

All 3 swerves ended in lane following, but each turn was near 130 degrees, not 45.

29 OCT · REBUILD

Turn 90, curve, turn 90

Turn rates measured: 305 deg/s, not 120. New moves tested 5 of 5 on the track.

Full entries: DECISIONS.md, AVOID2, MOVE2 and AVOID3. Both runs are filmed.

Inference and control loop (per-frame)

1 · CAMERA

Frame

224 by 224 pixels

2 · PERCEPTION

Three models

ResNet18, TensorRT

```
lane: visible, x
curve: class, side
cup: row, x
```

3 · DECISION

State machine

Six states, priority order

4 · CONTROL

PD and speed

Gains scaled by speed, curve feedforward

5 · MOTORS

Wheels

Battery, camera and motor watchdogs

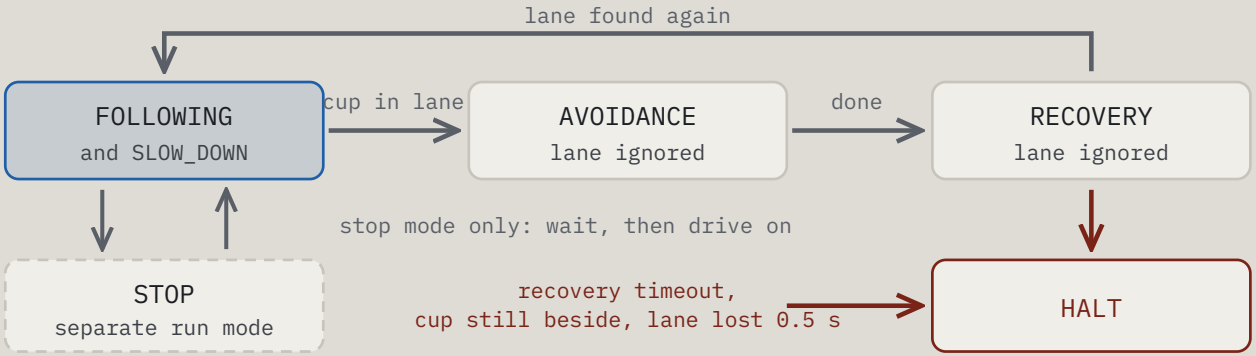
- Three ResNet models provide lateral coordinates and classification.
- Color masks calculate exact physical geometry for safety-critical decisions.

THE LOOP, IN FIVE LINES

Loop measured in the Task 1 runs. TensorRT took one lane pass from 41 ms to 10.6 ms.

Six states, priority settles conflicts

DRIVING ACROSS THE TOP, STOP MODE AND THE ENDING BELOW



Order: **halt**, **avoidance** and **recovery**, **stop**, **slow down**, **following**. During the swerve the lane rule is off on purpose.

THE BRANCHES THAT ARE THE PRIORITY

Perception model performance and known limitations

lane

99.6 %

frames where it finds the lane · center off by 0.022 · 2458 training frames

Single row lacks heading angle and cannot differentiate curve entry from lateral offset.

curve

88.6 %

five classes correct · side of all 37 sharp frames right · 3276 frames

6 of 37 sharp turns misclassified as gentle, causing delayed feedforward steering.

cup

98.1 %

cup seen or not · distance off by 8 rows · 3147 frames

Zero false positives on 454 clean frames. 11 misses occurred strictly at image borders.

Measured on whole laps held out of training. Labels come from a color mask, so these are agreement with that mask.
models/*.json

Task 1: our control layer drove 60 s both ways



Seconds driven against a 60 s cap. Red ended in a lane loss. The curve model bought the sharp curve, not smoothness.

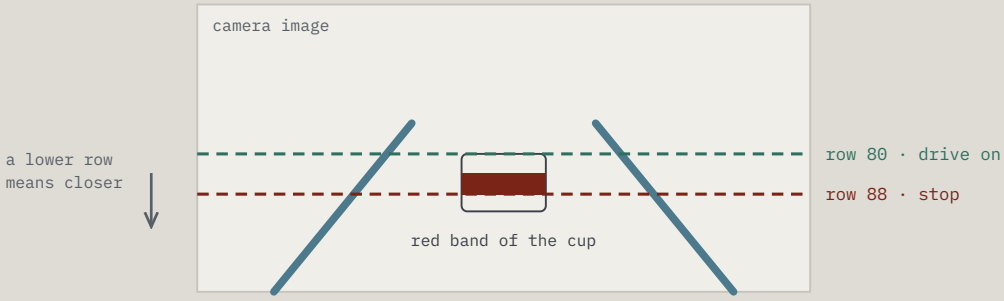
WHY THE STEERING HAS THREE TERMS

Six runs, 20 and 21 September. Wobble: stock 0.150 and 0.126, our PD 0.087, adaptive 0.099. Refreshing the table with today's code is an open step.

Obstacle detection and distance-gated stop

8 STOPS, SHOWN LIVE

WHAT THE ROBOT SEES · ROWS OF THE 224 PIXEL IMAGE



- Stop condition: obstacle band reaches row >= 88 for four consecutive frames.
- Hysteresis release: drives on when clear above row 80 to prevent chatter.

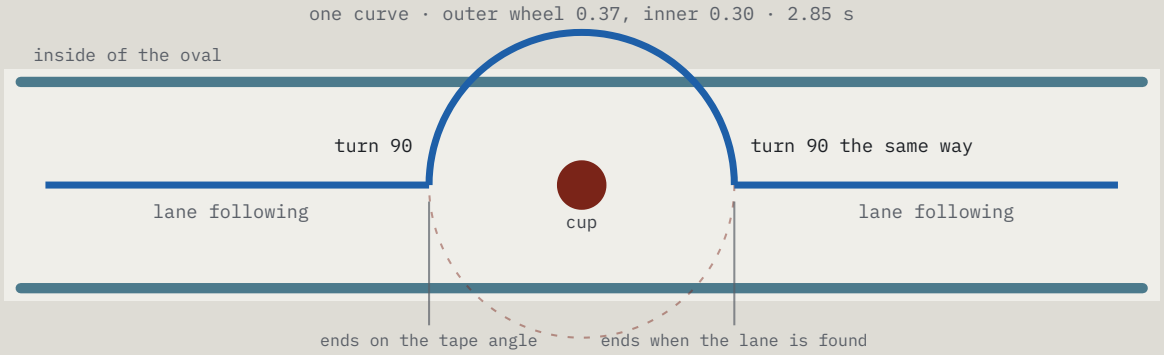
THE TRIGGER, IN FOUR LINES

Run demo_stop_4, 90 s, no lane loss. Five of eight stops settled at rows 90 to 104, which the distance survey puts at roughly 25 to 40 cm. A person lifted each cup away.

Obstacle circumvention and lane recovery

TESTED WITH AND WITHOUT A CUP

THE INTENDED PATH · TOP VIEW, INSIDE OF THE OVAL



- Trajectory: 90° pivot off lane → constant-radius bypass curve (2.85 s) → 90° return pivot.
- Track test results: 5 of 5 maneuvers achieved target 90° rotation without obstacle contact.

turn_test.py and swerve_test.py. In the tests the robot finished near the outer tape rather than the center, so the width of the curve is still being tuned.

FAILURES AND THEIR CAUSES

Root cause analysis & anomaly resolutions

WHAT WE SAW	CAUSE	FIX, AND IT HELD
Left the lane in the sharp curve	The turn direction came from the lane center, which sits near zero there and flips sign	The curve model learned five classes and reports the side. 60 s both ways
OOM crash & GStreamer camera buffer stall	Training ran while driving, memory ran out, the camera died and kept returning one frozen frame	Stop after 0.5 s without a new frame. It caught a real freeze at 46.6 s later
Open-loop yaw overshoot (≈ 130° vs. 45° nominal) due to uncalibrated angular rate (305 deg/s vs. 120 deg/s)	Pivots timed from 120 deg/s; actual rate is 305 deg/s, causing substantial turn overshoot	Turns end on the tape angle in the camera, not on a timer
Steering got jerkier with unchanged code	Not found. An A/B run with the old commit swung just as much, so it is the robot or the track, not the code	STILL OPEN: A motor trim and a smoothed D term are under test. This is our question for today

Each one written up as problem, hypothesis, change, test, result: DECISIONS.md entries DATA2, CURVE3, MOVE2, AB1, STEER2.

NEXT

What we do before the next demonstration

☐	Put the three moves into the state machine	pass: recorded frames give the same state through old and new code
☐	Repeat the avoidance ten times, filmed	pass: 8 of 10 passes end in lane following, untouched
☐	Refresh the Task 1 table	pass: the six runs repeated with today's code
☐	Team: tape an intersection and a parking box	blocks tasks 3, 4 and 5. Lane and tape width still unmeasured

The last row is the only one that blocks other tasks. Signs, intersection and parking all reuse the pipeline that already works, so the code is not the bottleneck.

Steps and pass conditions from PLAN.md section 6. Escape returns to the roadmap, d gives the document version.